

Module 1: reduced resistive-MHD state

The project physics is a strict two-dimensional, incompressible, doubly periodic reduced-MHD model:

$$\Omega = [0, 2\pi) \times [0, 2\pi), \quad t \in [0, T], \quad \partial_z = 0.$$

The primary scalar fields are magnetic flux and stream function,

$$\mathbf{q}(x, y, t) = (\psi, U),$$

with derived quantities

$$\mathbf{B}_\perp = \nabla\psi \times \hat{\mathbf{z}} = (\psi_y, -\psi_x), \quad \mathbf{v}_\perp = \hat{\mathbf{z}} \times \nabla U = (-U_y, U_x),$$

$$J = -\nabla^2\psi, \quad \omega = \nabla^2 U, \quad \nabla \cdot \mathbf{B}_\perp = \nabla \cdot \mathbf{v}_\perp = 0.$$

The revised learning direction uses the same physics, but the surrogate input becomes a complete initial field pair rather than only coordinates and scalar case parameters.

Module 1: brackets and governing equations

The two-dimensional bracket is

$$[f, g] = f_x g_y - f_y g_x, \quad \mathbf{v}_\perp \cdot \nabla f = [U, f].$$

Thus nonlinear advection appears compactly as a Poisson bracket. The frozen sign convention is

$$\boxed{\psi_t + [U, \psi] = \eta \nabla^2 \psi,}$$

$$\boxed{\omega_t + [U, \omega] = [J, \psi] + \nu \nabla^2 \omega,} \quad J = -\nabla^2 \psi, \quad \omega = \nabla^2 U.$$

Interpretation:

$[U, \psi]$: flux advection, $\eta \nabla^2 \psi = -\eta J$: resistive slippage,

$[J, \psi]$: curl of Lorentz force, $\nu \nabla^2 \omega$: viscous vorticity diffusion.

Module 1: scales, gauges, and energy

Alfven normalization gives

$$V_A = \frac{B_0}{\sqrt{\mu_0 \rho_0}}, \quad t_A = \frac{L_0}{V_A}, \quad \eta = \frac{\eta_m}{L_0 V_A}, \quad \nu = \frac{\nu_d}{L_0 V_A}.$$

Dimensionless control numbers are

$$S = \eta^{-1}, \quad \text{Re} = \nu^{-1}, \quad \text{Pm} = \frac{\nu}{\eta}.$$

Periodic gauges and means:

$$\langle U \rangle(t) = 0, \quad \frac{d}{dt} \langle \psi \rangle = 0, \quad \langle J \rangle = \langle \omega \rangle = 0.$$

The verification energy law is

$$\frac{d}{dt} \frac{1}{2} \int_{\Omega} (|\nabla \psi|^2 + |\nabla U|^2) dA = -\eta \int_{\Omega} J^2 dA - \nu \int_{\Omega} \omega^2 dA \leq 0.$$

Module 1: claim boundary under the operator change

The physical model does *not* change when the neural architecture changes. What changes is the learned map:

old coordinate surrogate: $(x, y, t, \eta, \nu, A_0) \mapsto (\psi, U),$

new full-field operator: $(\psi_0, U_0, \eta, \nu, t) \mapsto (\psi_t, U_t).$

In Module 1 language, this means:

(ψ_0, U_0) carries the initial topology and flow shape, (η, ν) carry transport strength.

A_0 may remain metadata, but in version 1 it is already encoded through U_0 .

The scientific boundary remains strict: these notes support a synthetic two-field reduced-MHD surrogate workflow, not a validated surrogate of genuine M3D-C1 output.

Module 2: solver output becomes operator data

For each physical case

$$\mu = (\eta, \nu, A_0),$$

the verified solver produces a full trajectory

$$\mathcal{Q}_\mu = \{\psi(x, y, t), U(x, y, t), J(x, y, t), \omega(x, y, t) : (x, y) \in \Omega, 0 \leq t \leq T\}.$$

The integrated revision makes the downstream handoff explicit:

$$X_k = (\psi[0, :, :], U[0, :, :], \eta, \nu, t_k), \quad Y_k = (\psi[k, :, :], U[k, :, :]).$$

These are raw numerical scalar arrays. Contours, field-line plots, PNGs, screenshots, and rendered images are excluded from the learning-data path.

Module 2: island coalescence initial condition

The benchmark starts from

$$\psi_0(x, y) = \Psi_{\text{eq}} \sin x \sin y, \quad \Psi_{\text{eq}} = 1,$$

$$U_0(x, y) = A_0(\cos x - \cos y).$$

Derived initial fields are

$$J_0 = -\nabla^2 \psi_0 = 2\psi_0, \quad \omega_0 = \nabla^2 U_0 = -U_0,$$

$$\mathbf{B}_{\perp,0} = (\sin x \cos y, -\cos x \sin y), \quad \mathbf{v}_{\perp,0} = (-A_0 \sin y, -A_0 \sin x).$$

Important limitation for operator learning:

$$U_0^{(A_0)}(x, y) = A_0 f(x, y) \quad \Rightarrow \quad \text{one fixed initial-flow shape with varying amplitude.}$$

Module 2: pseudo-spectral reference calculation

The solver advances the Fourier state

$$\hat{\mathbf{q}}(t) = \begin{bmatrix} \hat{\psi}(t) \\ \hat{\omega}(t) \end{bmatrix}, \quad \hat{U} = -\frac{\hat{\omega}}{k^2} \ (k \neq 0), \quad \hat{J} = k^2 \hat{\psi}.$$

Spectral derivatives are algebraic:

$$\hat{f}_x = ik_x \hat{f}, \quad \hat{f}_y = ik_y \hat{f}, \quad \widehat{\nabla^2 f} = -k^2 \hat{f}.$$

Nonlinear brackets are dealiased physical-grid products:

$$\widehat{\mathcal{B}(f, g)} = \mathcal{P}_{2/3} \mathcal{F}(f_x g_y - f_y g_x), \quad |m| \leq \lfloor N_x/3 \rfloor, \quad |n| \leq \lfloor N_y/3 \rfloor.$$

Classical RK4 advances the stage-consistent state; (U, J) are reconstructed at every stage, not reused from stale fields.

Module 2: acceptance before pair extraction

A trajectory is accepted only after verification and numerical adequacy:

operators $\rightarrow$ analytic fields $\rightarrow$ brackets $\rightarrow$ exact decay

$\rightarrow$ manufactured solution $\rightarrow$ refinement and parity.

Physics diagnostics include

$$E = \frac{1}{2} \int_{\Omega} (|\nabla \psi|^2 + |\nabla U|^2) dA, \quad D_{\eta} = \eta \int_{\Omega} J^2 dA, \quad D_{\nu} = \nu \int_{\Omega} \omega^2 dA,$$

$J_{\max}$, sheet width, spectral tails, $\Delta\psi_{OX} = \psi_X - \psi_O$, $\Phi_{\text{rec}} = \Delta\psi_{OX}(0) - \Delta\psi_{OX}(t)$.

One trajectory may yield many target-time operator pairs, but it is still one physical case.

Splits must occur before extracting $X_k \mapsto Y_k$ pairs.

Module 3: full-field operator example

Module 3 defines the learning object without modifying canonical solver truth:

$$\mathcal{X}_{c,k} = (\psi_{c,0}, U_{c,0}, \eta_c, \nu_c, t_k), \quad \mathcal{Y}_{c,k} = (\psi_{c,k}, U_{c,k}).$$

Tensor meanings are explicit:

$$\psi_{c,k}, U_{c,k} \in \mathbb{R}^{N_x \times N_y}, \quad q[n, i, j] = q(t_n, x_i, y_j).$$

Identifiers must remain separate:

$$\text{physical_case_id} \neq \text{case_id} \neq \text{initial_state_id} \neq \text{operator_example_id}.$$

Different times from one trajectory are correlated examples, not independent cases.

Module 3: HDF5 contract and tensor packing

Canonical case files store fields before learning views:

$$\text{fields/psi}, \text{fields/U}, \text{fields/J}, \text{fields/omega} \in \mathbb{R}^{N_t \times N_x \times N_y}.$$

The operator tensor view may pack channels as

$$\begin{aligned} \mathcal{X}_{c,k} &\in \mathbb{R}^{C_{\text{in}} \times N_x \times N_y}, \quad C_{\text{in}} = 5 : \quad [\psi_0, U_0, \eta \mathbf{1}, \nu \mathbf{1}, t_k \mathbf{1}], \\ \mathcal{Y}_{c,k} &\in \mathbb{R}^{2 \times N_x \times N_y} : \quad [\psi_k, U_k]. \end{aligned}$$

Broadcast scalar channels are constant fields. No image resizing, color mapping, rasterization, or aspect-ratio distortion is permitted.

Module 3: splitting, scaling, and leakage

Splits are made on complete physical trajectories:

$$\mathcal{P}_{\text{all}} = \mathcal{P}_{\text{tr}} \cup \mathcal{P}_{\text{va}} \cup \mathcal{P}_{\text{ti}} \cup \mathcal{P}_{\text{te}}, \quad \mathcal{P}_a \cap \mathcal{P}_b = \emptyset.$$

Only after this step are target times selected:

$$(c \in \mathcal{P}_s) \Rightarrow (c, k) \in \mathcal{E}_s \quad \text{for all allowed } k.$$

Training-only transforms are fitted once:

$$(m_\psi, s_\psi, m_U, s_U, m_\eta, s_\eta, m_\nu, s_\nu) = \text{fit}(\mathcal{P}_{\text{tr}}).$$

For case-balanced sampling,

$$\Pr(c) = \frac{1}{|\mathcal{P}_{\text{tr}}|}, \quad \Pr(k \mid c) = \text{declared target-time rule}.$$

Random pair splits would leak nearly identical time-adjacent physics into test sets.

Module 3: M3D-C1-shaped I/O without false equivalence

The dataset is M3D-C1-shaped only at the interface level:

coordinates + *extfields* + *extparameters* + *extdiagnostics* + *extmetadata* + splits.

It is not genuine M3D-C1 physics. An eventual adapter must declare

geometry, units, mesh, field names, signs, gauges, closures, projection losses.

Module 3 release checks include

axis parity,

round trip, $J + \nabla^2 \psi$,

$\omega - \nabla^2 U$,

$\langle U \rangle$, split disjointness,

no-rendering provenance.

The learning view is reversible bookkeeping; it must not overwrite accepted reference arrays.

Module 4: data-only neural-operator baseline

The baseline is no longer a coordinate MLP. It is a full-field neural operator:

$$\mathcal{G}_\theta : (\psi_0, U_0, \eta, \nu, t) \mapsto (\hat{\psi}_t, \hat{U}_t).$$

Inputs and outputs are numerical arrays:

$$\psi_0, U_0, \hat{\psi}_t, \hat{U}_t \in \mathbb{R}^{N_x \times N_y}.$$

A strict data-only baseline answers:

How well can observed trajectory pairs alone train a field-to-field surrogate?

No PDE residual, initial-condition penalty, boundary loss, energy balance, target J , target ω , derivative loss, or spectral loss updates the weights.

Module 4: Fourier neural operator structure

A version-1 input tensor can be

$$\mathcal{X} = [\psi_0, U_0, \eta \mathbf{1}, \nu \mathbf{1}, t \mathbf{1}] \in \mathbb{R}^{5 \times N_x \times N_y}.$$

FNO layers alternate local channel mixing and truncated spectral convolution:

$$\mathbf{h}^{(\ell+1)}(x, y) = \sigma(W^{(\ell)} \mathbf{h}^{(\ell)}(x, y) + \mathcal{F}^{-1}[R^{(\ell)}(k) \mathcal{F} \mathbf{h}^{(\ell)}(k)](x, y)).$$

The baseline architecture is a scientific comparator, not the final claim:

lift $\rightarrow$ 4 FNO blocks $\rightarrow$ projection to (ψ, U) ,

with width, retained modes, activation, and depth selected only by validation cases.

Module 4: strict supervised full-field loss

For one operator pair (c, k) , standardized output errors are fields:

$$E_{\psi,c,k} = \hat{\tilde{\psi}}_{c,k} - \tilde{\psi}_{c,k}, \quad E_{U,c,k} = \hat{\tilde{U}}_{c,k} - \tilde{U}_{c,k}.$$

The case-balanced data-only loss is

$$\mathcal{L}_{\text{data}} = \frac{1}{|\mathcal{C}_B|} \sum_{c \in \mathcal{C}_B} \frac{1}{|\mathcal{K}_c|} \sum_{k \in \mathcal{K}_c} \left[\frac{\|E_{\psi,c,k}\|_2^2}{N_x N_y} + \frac{\|E_{U,c,k}\|_2^2}{N_x N_y} \right].$$

Training-time counts must distinguish

#trajectories $\neq$ #operator pairs $\neq$ #pixels/grid points.

Large numbers of target times do not create new independent physical cases.

Module 4: evaluation and limitation

Full-field metrics use inverse-transformed predictions:

$$\epsilon_{L_2}^{(q)}(c, k) = \frac{\|\hat{q}_{c,k} - q_{c,k}\|_2}{\|q_{c,k}\|_2 + \epsilon}, \quad q \in \{\psi, U, J, \omega\}.$$

Derived diagnostics are computed after prediction:

$$\hat{J} = -\nabla^2 \hat{\psi}, \quad \hat{\omega} = \nabla^2 \hat{U}, \quad E_B, E_K, J_{\max}, \Phi_{\text{rec}}, \text{ spectra, O/X points.}$$

Evaluation-only residuals may be reported, but not trained:

$$R_\psi, \quad R_\omega \quad \text{are diagnostics in Module 4.}$$

Limitation: because $U_0 = A_0(\cos x - \cos y)$, the current dataset tests operator learning over transport coefficients, time, and amplitude, not arbitrary initial-state shape generalization.

Module 5: physics-informed neural operator

Module 5 keeps the same full-field FNO map as Module 4:

$$\mathcal{G}_\theta : (\psi_0, U_0, \eta, \nu, t) \mapsto (\hat{\psi}_t, \hat{U}_t).$$

The model becomes physics-informed when reduced-MHD residuals affect training. The derivative strategy is

$\text{AD in scalar time } t \quad + \quad \text{Fourier spectral differentiation in } (x, y).$

Predicted derived fields are

$$\hat{J} = -\nabla^2 \hat{\psi}, \quad \hat{\omega} = \nabla^2 \hat{U}.$$

This is a physics-informed neural operator/PINO formulation, not the old coordinate-MLP PINN.

Module 5: full-field residual equations

For each physics context $(\psi_0, U_0, \eta, \nu, t)$, form complete residual fields:

$$R_\psi = \hat{\psi}_t + [\hat{U}, \hat{\psi}] - \eta \nabla^2 \hat{\psi},$$

$$R_\omega = \hat{\omega}_t + [\hat{U}, \hat{\omega}] - [\hat{J}, \hat{\psi}] - \nu \nabla^2 \hat{\omega}.$$

Spatial operations use the same periodic spectral conventions as the Module 2 solver:

$$[f, g] = f_x g_y - f_y g_x, \quad \widehat{\mathcal{B}(f, g)} = \mathcal{P}_{2/3} \mathcal{F}(f_x g_y - f_y g_x).$$

The vorticity residual still contains $\nabla^4 \hat{U}$ mathematically, but it is computed by spectral multiplication rather than four nested coordinate-AD passes.

Module 5: objective and constraints

The primary loss is

$$\mathcal{L}_{\text{total}} = w_d \mathcal{L}_{\text{data}} + w_f \mathcal{L}_{\text{PDE}} + w_0 \mathcal{L}_{\text{IC}} + w_g \mathcal{L}_{\text{gauge}}, \quad \mathcal{L}_{\text{BC}} = 0.$$

The PDE term is an equation-balanced field norm:

$$\mathcal{L}_{\text{PDE}} = \lambda_\psi \frac{\|R_\psi/S_\psi\|_2^2}{N_x N_y} + \lambda_\omega \frac{\|R_\omega/S_\omega\|_2^2}{N_x N_y}.$$

The operator identity at the initial time may be imposed by

$$\mathcal{L}_{\text{IC}} = \|\hat{\psi}_{t=0} - \psi_0\|^2 + \|\hat{U}_{t=0} - U_0\|^2.$$

The stream-function gauge remains

$$\mathcal{L}_{\text{gauge}} \propto \langle \hat{U} \rangle^2.$$

Energy-balance losses are controlled ablations, not silently added to the base model.

Module 5: controlled evidence that physics helped

The comparison with Module 4 must hold fixed

trajectories, target times, preprocessing, FNO capacity, $\mathcal{L}_{\text{data}}$, test cases, metrics.

Physics helps only if it improves declared outcomes:

$\epsilon_{\psi,U}$, $\epsilon_{J,\omega}$, $\|R_{\psi}\|$, $\|R_{\omega}\|$, E balance, $J_{\max}$, Φ_{rec} , event timing, data efficiency.

Failure modes remain plausible:

loss imbalance, gradient conflict, oversmoothing sheets,
aliasing mismatch, test-case leakage, false extrapolation confidence.

Scientific boundary: the result is still a surrogate of verified synthetic two-field reduced-MHD data. M3D-C1 fidelity requires genuine M3D-C1 arrays and renewed validation.